

TW09: Statistical Thinking for Data Science

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Use descriptive statistics, probability, distributions, correlation, and hypothesis testing to produce a cautious evidence-based memo.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Build a statistical variable audit

Scenario

A manager has supplied a small customer table but has not identified which summaries are valid for each field.

INSTRUCTIONS

1. Create a data dictionary for customer_id, region, satisfaction band, order count, waiting time, and spend.
2. Classify every variable as nominal, ordinal, discrete, or continuous.
3. Choose one suitable summary and one unsuitable operation for each variable.
4. Explain why an identifier containing digits is not automatically numerical.

REQUIRED SUBMISSION

- A completed six-row variable audit.
- A short interpretation of two classifications that could easily be mistaken.
- At least two programmatic checks of the supplied table.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Describe a skewed business measure

Scenario

The finance team wants a typical salary figure, but one executive value is much larger than the others.

INSTRUCTIONS

1. Calculate count, mean, median, mode, range, variance, standard deviation, Q1, Q3, and IQR.
2. Apply the 1.5 x IQR rule and list flagged observations.
3. Repeat mean and median after excluding flagged values for sensitivity only.
4. Recommend a measure of centre without automatically deleting the outlier.

REQUIRED SUBMISSION

- A labelled descriptive-statistics table.
- An outlier flag generated from calculated fences.
- A four-to-six sentence recommendation discussing robustness and context.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Simulate and verify probability

Scenario

A quality process succeeds with probability 0.65 on each independent attempt.

INSTRUCTIONS

1. Simulate 20,000 sets of 12 attempts using random seed 123.
2. Estimate the probability of exactly 8 successes and at least 8 successes.
3. Calculate both probabilities theoretically with a binomial model.
4. Report absolute simulation error and explain why simulation and theory differ slightly.

REQUIRED SUBMISSION

- Reproducible simulation code.
- A comparison table containing simulated, theoretical, and absolute-error values.
- A correct statement of binomial assumptions.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Compare Poisson and normal questions

Scenario

A support desk averages 4.2 calls per hour, while delivery times are approximately normal with mean 32 minutes and standard deviation 6 minutes.

INSTRUCTIONS

1. Calculate the probability of exactly 5 calls and more than 6 calls.
2. Calculate the delivery-time z-score for 44 minutes.
3. Calculate the probability of a delivery between 26 and 38 minutes.
4. Explain why a Poisson PMF and a normal PDF are not interpreted in the same way.

REQUIRED SUBMISSION

- Four correctly labelled calculations.
- A distribution-choice justification for each variable.
- A short explanation distinguishing PMF, PDF, and CDF.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Write a correlation and hypothesis-test memo

Scenario

Two independently sampled service teams have completion scores, and management also wants to know whether training hours are linearly related to scores.

INSTRUCTIONS

1. Calculate Pearson correlation and its p-value for paired training hours and scores.
2. Check the paired values visually with a scatter plot.
3. Run a Welch independent t-test for Team A versus Team B.
4. Report group means, mean difference, test statistic, p-value, and a 0.05 decision.
5. Write a memo that separates association, statistical significance, effect size, practical importance, and causation.

REQUIRED SUBMISSION

- One scatter plot and a complete statistical results table.
- Correct null and alternative hypotheses.
- A 150-to-200-word evidence memo with no overstated claim.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____